engineering 4 us all

Student Handout

Name: Ninja Cat **Date:** 11/11

Element C: Presentation and justification of solution design requirements

1. Design Requirement for "Handle" Ergonomics (High Priority)

The cart solution must feature a pushing mechanism that allows pushing without bending over and minimizes back strain by supporting a neutral spine position. The cart solution must feature a handle that allows pushing at a height adjustable between 3 to 5 feet to support users of various heights. The handle must also keep the user's wrist angle within ~10 degrees of neutral and allow pushing with less than ~15 pounds of horizontal force. This addresses Mr. Kiser's need to avoid arching and twisting the back, leading to discomfort due to prior slipped discs.

2. Design Requirement for Durability (High Priority)

The cart solution must be designed to withstand at least five years of regular use, supporting up to 500 pounds across both cart levels. The cart solution must withstand at least 10,000 cycles of movement on a treadmill (representing approximately five years of regular use) and maintain structural integrity while supporting a load of 500 pounds. This requirement aligns with durability expectations for a long-term, reliable solution.

3. Design Requirement for Removability (High Priority)

The cart modification should be easily removable without needing tools to accommodate different users or allow cart use in its original form. The cart modification must be installable or removable in less than 2 minutes, ideally without requiring tools, demonstrated through usability testing by at least 10 users with an 80% success rate. This allows for a solution that doesn't permanently alter the cart and can be shared among multiple users.

4. Design Requirement for Battery and Range Indicator (Medium Priority)

If motorized, the cart solution should include a battery indicator and provide sufficient range for multiple trips (2-3 hallways) on a single charge. Access to

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motorization provided the battery can handle frequent usage and has a visible charge indicator so many trips can be made.

5. Design Requirement for Compact Design (Medium Priority)

The cart modification should remain compact, with any added components staying close to the cart frame and minimally encroaching on loading space. The cart modification must not extend more than 1 foot beyond the existing cart frame on any side and must retain at least 90% of the original loading space, verified through dimensional analysis. It ensures the modified solution does not hinder loading and maintains existing cart utility.

6. Design Requirement for Push-Only Functionality (Medium Priority)

The modification should enable pushing the cart easily without requiring a pulling motion, which might cause further back strain. The modification must allow the cart to be moved forward with a horizontal force not exceeding ~15 pounds, tested under a fully loaded condition of 500 pounds, without requiring any pulling motion. Pushing is the primary movement that addresses Mr. Kiser's current back pain issues by eliminating twisting and arching motions.

7. Design Requirement for Handle Flexibility (Low Priority)

The modified cart handle must allow minimal flex to absorb motions without compromising control and force application. The modified cart handle must allow no more than ~2 inches of deflection under a force of 20 pounds applied at the handle's midpoint Minimizing motion aligns with Mr. Kiser's desire for reduced back strain and enhanced user comfort.

Digital Signature of Primary Stakeholder: